Acute Effect of Different Velocity-Based Training Protocols on 2000-meter Rowing Ergometer Performance

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Abstract

Leandro Quidel-Catrilelbún, ME, Ruiz-Alias, SA, García-Pinillos, F, Ramirez-Campillo, R, and Pérez-Castilla, A. Acute effect of different velocity-based training protocols on 2000-m rowing ergometer performance. *J Strength Cond Res* 38(1): e8–e15, 2024—This study aimed to explore the acute effect of 4 velocity-based resistance training (VBT) protocols on 2000-m rowing ergometer (RE_{2000}) time trial, as well as the behavior of the maximal neuromuscular capacities when RE_{2000} is performed alone or preceded by VBT protocols in the same session. Fifteen male competitive rowers (15–22 years) undertook 5 randomized protocols in separate occasions: (a) RE_{2000} alone (control condition); (b) VBT against 60% of 1 repetition maximum (1RM) with a velocity loss in the set of 10% followed by RE_{2000} ($VBT_{60\text{-}10}$ + RE_{2000}); (c) VBT against 60% 1RM with a velocity loss in the set of 30% followed by RE_{2000} ($VBT_{60\text{-}30}$ + RE_{2000}); (d) VBT against 80% 1RM with a velocity loss in the set of 10% followed by RE_{2000} ($VBT_{80\text{-}10}$ + RE_{2000}); (e) VBT against 80% 1RM with a velocity loss in the set of 30% followed by RE_{2000} ($VBT_{80\text{-}30}$ + RE_{2000}). The load-velocity relationship (load-axis intercept [L_0], velocity-axis intercept [v_0], and area under the load-velocity relationship line [A_{line}]) was used to evaluate the maximal neuromuscular capacities during the prone bench pull exercise before and after each protocol. The time trial was significantly longer for $VBT_{60\text{-}30}$ + RE_{2000} and $VBT_{80\text{-}30}$ + RE_{2000} than for RE_{2000}, $VBT_{60\text{-}10}$ + RE_{2000} and $VBT_{80\text{-}10}$ + RE_{2000} (all $p < 0.001$; ES = 0.10–0.15). L_0 and A_{line} were significantly reduced after all protocols ($p < 0.001$; ES = 0.10–0.13), with A_{line} reduction more accentuated for $VBT_{60\text{-}10}$ + RE_{2000}, $VBT_{60\text{-}30}$ + RE_{2000}, $VBT_{80\text{-}30}$ + RE_{2000}, and RE_{2000} (all $p = 0.001$; ES = 0.11–0.18) than for $VBT_{80\text{-}10}$ + RE_{2000} ($p = 0.065$; ES = 0.05). Therefore, VBT protocols with greater velocity loss in the set (30% vs. 10%) negatively affected subsequent rowing ergometer performance, in line with impairment in A_{line} pulling performance.

Key Words: endurance training, human physical conditioning, musculoskeletal and neural, resistance training, water sports

Introduction

Rowing has received increasing attention in the strength and conditioning field (11,20). Briefly, rowing involves repeated stroking cycles to generate propulsive forces that drive the boat-rower system as far and fast as possible during a racecourse (17). A typical rowing competition takes place over a 2000-m course and lasts approximately 6 minutes (11). Rowing performance not only depends on aerobic metabolism but also on neuromuscular capacities such as maximal stroke force and power (16,19). Indeed, resistance and endurance training performed concurrently as part of a periodized training program (i.e., concurrent training) has optimized rowing performance (5,18). However, concurrent training prescription should be undertaken with care given that the resistance training–induced fatigue may impair the quality of the subsequent rowing endurance training sessions and, consequently, induce suboptimal endurance improvements (2). This phenomenon has been referred to as "resistance training-induced suboptimization on endurance performance" (RT-SEP) and may be modulated by several concurrent training variables (e.g., between-mode recovery period, order of training mode, exercise intensity) (3). Nevertheless, the acute effect of resistance training on rowing endurance performance is underresearched (10).

Among the concurrent training variables, exercise intensity and volume are probably the most critical in determining the type and extent of endurance response to resistance training (2). It has been suggested that high-intensity and high-volume resistance training may increase susceptibility to RT-SEP more than low-intensity and low-volume resistance training (3). Instead, Gallagher et al. (5) observed that high-load low-repetition resistance training tend to improve 2000-m rowing ergometer performance more than low-load high-repetition resistance training (15 vs. 12 seconds) in male varsity rowers. However, Ebben et al. (4) highlight the importance of considering the level of the rowers because those with a lower pretraining status are more prone to obtain greater benefits from low-

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Journal of Strength and Conditioning Research 38(1)/e8–e15

intensity training rather than high-intensity resistance training. Therefore, further study is needed to draw stronger conclusions on the topic. Of note, the methods commonly used to configure training stimuli (e.g., 1 repetition maximum [1RM]; maximal load that can be lifted a given number of repetitions) do not guarantee a homogeneous level of effort between individuals because of several potential limitations, discussed elsewhere (12).

In response to these limitations, velocity-based training (VBT) has emerged as a contemporary method that allows the optimal autoregulation and individualization of resistance training intensity and volume because of the strong relationship observed between the movement velocity and the percentage of 1RM (7,24) and the magnitude of velocity loss in the set and the percentage of performed repetitions against different relative intensities (12,31). Based on this novel methodology, Pareja-Blanco et al. (22) reported greater fatigue and a slower rate of recovery following a VBT protocol with a low relative load (60% 1RM) and high velocity loss (40% velocity) than with a high relative load (80% 1RM) and low velocity loss (20%). More specifically, Held et al. (13) observed that VBT with a 10% velocity loss improved strength capacity more than resistance training to failure in highly trained rowers during an 8-weeks concurrent training period. Sánchez-Moreno et al. (33) showed that an 8-week concurrent VBT period with 15% (and not 45%) velocity loss resulted in greater gains in maximal aerobic capacity than endurance training alone. However, the acute effect of different concurrent VBT protocols in terms of loading magnitude (e.g., 60% vs. 80% 1RM) and velocity loss in the set (e.g., 10% vs. 30% 1RM) on rowing performance measures is underresearched.

The quality of endurance training may be compromised when resistance training induced residual fatigue as a result of inappropriate exercise prescription (2,3). With VBT, the intensity and volume are adjusted to the athletes' readiness to train, thus better fatigue management (34). However, there is scarce of information about the mechanical fatigue induced by different concurrent VBT protocols (3). Nájera-Ferrer et al. (21) observed that regardless of the order between VBT and running training, compared with 20% velocity loss in the set, 40% velocity loss in the set induced higher mechanical stress (i.e., reduction in jump height and back squat velocity). Pérez-Castilla et al. (23) found that running sprint interval training deteriorates the resistance training quality (i.e., number of repetitions and velocity of the set) and maximal force capacity in a greater magnitude than long high-intensity interval training. It is important to note that Nájera-Ferrer et al. (21) assessed mechanical fatigue under a single mechanical condition (e.g., unloaded vertical jump), whereas Pérez-Castilla et al. (23) used the linear force-velocity modeling approach based only on 2 distant loads (referred to as "*2-point method*"). The latter approach provides a selective assessment of the maximal neuromuscular capacities, but the measurement accuracy of the maximal velocity and power capacities might be affected by the large extrapolation required from the experimental points to the velocity axis (25). Consequently, the variables derived from the load-velocity (L-V) relationship (load-axis intercept [L_0], velocity-axis intercept [v_0], and area under the load-velocity relationship line [A_{line}]) have recently been recommended to simply and accurately assess maximal neuromuscular capabilities (25,26,30). However, changes in the magnitude of L-V relationship variables (L_0, v_0, and A_{line}) following different concurrent VBT protocols are currently unknown.

To overcome the above limitations and knowledge gaps, the aims of this study were (a) to explore the immediate effect of 4 different VBT protocols in terms of loading magnitude (60% vs. 80% 1RM) and velocity loss in the set (10% vs. 30%) on 2000-m rowing ergometer (RE_{2000}) time trial and (b) to examine the acute impact of RE_{2000} alone or preceded by the 4 different VBT protocols ($VBT_{60\text{-}10}$, $VBT_{60\text{-}30}$, $VBT_{80\text{-}10}$, and $VBT_{80\text{-}30}$) in the same session on the maximal neuromuscular capacities (L_0, v_0, and A_{line}). We hypothesized that (a) a low relative load (60% 1RM) and a high velocity loss (30%) would result in an increased RE_{2000} time trial than a high relative load (80% 1RM) and a low velocity loss (10%) (22) and (b) a decrement in A_{line}, more predominant in v_0 than in L_0, would be expected after the most fatiguing protocols ($VBT_{60\text{-}30} + RE_{2000} > VBT_{80\text{-}30} + RE_{2000} > VBT_{60\text{-}10} + RE_{2000} > VBT_{80\text{-}10} + RE_{2000} > RE_{2000}$) (23).

Methods

Experimental Approach to the Problem

A randomized crossover design was used to explore the acute effect of 4 different VBT protocols ($VBT_{60\text{-}10}$, $VBT_{60\text{-}30}$, $VBT_{80\text{-}10}$, and $VBT_{80\text{-}30}$) on RE_{2000} time trial, as well as the behavior of the maximal neuromuscular capacities when RE_{2000} is performed alone or preceded by the 4 different VBT protocols in the same training session ($VBT_{60\text{-}10} + RE_{2000}$, $VBT_{60\text{-}30} + RE_{2000}$, $VBT_{80\text{-}10} + RE_{2000}$, and $VBT_{80\text{-}30} + RE_{2000}$). Subjects undertook 5 randomized protocols in sessions separated by 48–72 hours (Figure 1 for further details). The individualized L-V relationship was determined during the prone bench pull exercise through the 2-point method before and after each protocol to assess residual fatigue on L_0, v_0, and A_{line}. Subjects were required to avoid any strenuous exercise throughout the study. All sessions were performed at the same time of the day for each subject (±1 hour) and under similar environmental conditions (~22° C and ~60% humidity).

Subjects

Fifteen male competitive rowers (7 cadets, 7 juniors and 1 senior; age = 16.7 ± 1.7 years [15–22 years], body height = 1.74 ± 0.07 m, body mass = 67.8 ± 8.0 kg, prone bench pull 1RM = 71.9 ± 23.3 kg; data presented as means ± standard deviation [*SD*]) volunteered to take part in this study. All athletes had participated in national competitions, and 3 of them participated in international competitions with the National Rowing Federation. In addition, 8 athletes are members of the National Promises program developed by the Ministry of Sports. All athletes completed 5 or more weekly training sessions with at least 2 strength-oriented training sessions. The prone bench pull exercise was regularly performed in their resistance training programs. No physical limitations, health problems, or musculoskeletal injuries that could compromise testing were reported. After being informed about the research purpose and experimental procedures, the subjects signed a written informed consent form before participation. The subjects aged <18 years granted verbal assent and their legal guardians signed the written informed consent form. The study was conducted according to the Declaration of Helsinki and was approved by the University of Granada Institutional Review Board.

Procedures

Body height using a stadiometer (Avanutri 201, RJ, Brazil) and body mass using a digital balance (Ventus B-300 A, Santiago, Chile) were measured at the beginning of the first session. Each protocol began with the same warm-up, which consisted of 5 minutes of rowing ergometer at a Pace Boat of about 140 seconds per 500 m, full-body dynamic stretching exercises, and 1 set of 6,

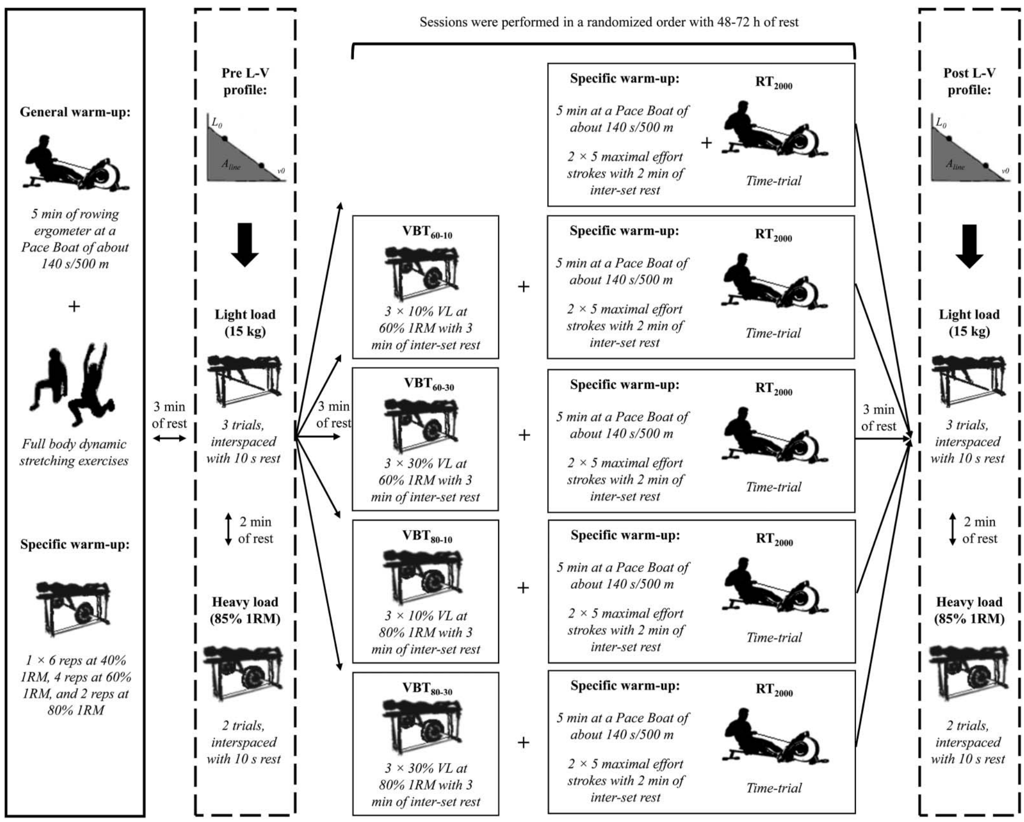

Figure 1. Overview of the experimental design. RE_{2000} = 2000-m rowing ergometer protocol; $VBT_{60\text{-}10}$ = velocity-based training (VBT) protocol against 60% of 1 repetition maximum (1RM) with a velocity loss (VL) in the set of 10%; $VBT_{60\text{-}30}$ = VBT protocol against 60% of 1RM with a VL in the set of 30%; $VBT_{80\text{-}10}$ = VBT against 80% of 1RM with a VL in the set of 10%; $VBT_{80\text{-}30}$ = VBT protocol against 80% of 1RM with a VL in the set of 30%; L-V profile = load-velocity profile; L_0 = load-axis intercept; v_0 = velocity-axis intercept; A_{line} = area under the L-V relationship line.

4, and 2 repetitions at 40, 60, and 80% of subjects' self-perceived prone bench pull 1RM, respectively. After 3 minutes of passive rest, and before the beginning of each protocol, the L-V relationship was determined during the prone bench pull exercise through the 2-point method (pre L-V profile) (23). Three trials against the 15-kg load (light) and 2 trials against the estimated 85% 1RM load (heavy; 61.2 ± 19.2 kg) were performed separately by 2 minutes of rest. The average score of the mean velocity values collected under the 2 different loading conditions was used for modeling the individualized L-V relationship (26). Specifically, the L-V relationship was determined using a least-square linear regression model: $L(V) = L_0 - s \times V$, in which L_0 represents the theoretical load at 0 velocity (i.e., load-axis intercept) and s is the slope of the L-V relationship (25,30). The v_0 (i.e., velocity-axis intercept) and A_{line} (i.e., area under the L-V relationship line) were then calculated as follows: $v_0 = L_0/s$ and $A_{line} = L_0 \times v_0/2$ (25). A linear velocity transducer (T-Force system; Ergotech, Murcia, Spain) was used to collect mean velocity values throughout the study (28). Subjects rested passively for 3 minutes between the completion of the pre L-V profile and the beginning of each protocol. The L-V relationship was also determined 3 minutes after completing each protocol (post L-V profile) (Figure 1).

RE_{2000} Protocol. The specific warm-up consisted of 5 minutes of rowing ergometer at a Pace Boat of about 140 seconds per 500 m, followed by 2 of 5 maximal effort strokes with 2 minutes of interset rest. After warming up, subjects performed the RE_{2000} protocol with the instruction to complete the 2000-m distance in the shortest time possible. A fixed-foot stretcher air-braked Concept2 Model D ergometer equipped with a PM5 monitor (Concept2 Morrisville, VT) was used for the RE_{2000} protocol. The resistance lever on the rowing machine flywheel was set at 120 (no units). The time trial was used as RE_{2000} performance indicator.

Velocity-Based Resistance Training Protocols. Two different relative loads (60% vs. 80% 1RM) and 2 different magnitudes of velocity loss during the set (10% vs. 30%) were used. Relative loads were determined from the individualized L-V relationship using the 3 warm-up sets (6). Briefly, the prone bench pull 1RM was estimated as the load associated with a mean velocity of 0.50 m·second^{-1} (34). Sets were terminated when (a) the corresponding target velocity loss limit was exceeded for 2 consecutive repetitions or (b) the subject is unable to complete 2 consecutive repetitions with a full range of motion (i.e., the barbell did not contact the underside of the bench). The fastest repetition on the first set was used to define the target velocity loss limit (e.g., if the fastest velocity is 1.00 m·second^{-1}, the target velocity used to guide set termination would be 0.90 m·second^{-1} or 0.70 m·second^{-1} for the 10% or 30% velocity loss, respectively). Specifically, the configuration of the 4 VBT protocols was as follows: (a) 60% 1RM with a velocity loss in the set of 10% ($VBT_{60\text{-}10}$), (b) 60% 1RM with a velocity loss in the set of 30% ($VBT_{60\text{-}30}$), (c)

80% 1RM with a velocity loss in the set of 10% ($VBT_{80\text{-}10}$), and (d) 80% 1RM with a velocity loss in the set of 30% ($VBT_{80\text{-}30}$). The same exercise (prone bench pull), number of sets (3), and interset rest duration (3 minutes) were used in all VBT protocols. The VBT performance indicators involved (1) load lifted, (2) number of repetitions completed, (c) fastest velocity of the set, and (d) average velocity of the set.

The bench pull technique involved the subjects lying down in a prone position, the chin in contact with the bench, elbows fully extended, and a self-selected prone grip of the barbell slightly wider than shoulder width (6). From that position, subjects lowered the barbell in a controlled manner until it contacts with a lower bench, held this position for approximately 2 seconds, and then pulled the barbell as fast as possible until it contacted the underside of the upper bench. The range of motion was individually measured and kept constant throughout all repetitions by adjusting the position of the lower bench. There was 8 cm between the underside of the bench and the subjects' chest. The legs were held during all repetitions with a rigid strap on the calves. Subjects received auditory mean velocity feedback immediately after completing each repetition to encourage maximal effort.

Statistical Analyses

Descriptive data are presented as mean ± *SD*s. The Shapiro-Wilk test confirmed the normal distribution of all variables ($p > 0.05$) except for the RE_{2000} time trial. A 1-way repeated-measures analysis of variance (ANOVA) was applied on the absolute load ($VBT_{60\text{-}10}$, $VBT_{60\text{-}30}$, $VBT_{80\text{-}10}$, and $VBT_{80\text{-}30}$) and pre L-V profile (RT_{2000}, $VBT_{60\text{-}10}$ + RT_{2000}, $VBT_{60\text{-}30}$ + RT_{2000}, $VBT_{80\text{-}10}$ + RT_{2000}, and $VBT_{80\text{-}30}$ + RT_{2000}). A 2-way repeated-measures ANOVA (protocol [$VBT_{60\text{-}10}$, $VBT_{60\text{-}30}$, $VBT_{80\text{-}10}$, and $VBT_{80\text{-}30}$] × set [first, second, and third]) was conducted on each VBT performance indicator (numbers of repetitions, fastest velocity, and average velocity). A 2-way repeated-measures ANOVA (protocol [RE_{2000}, $VBT_{60\text{-}10}$ + RE_{2000}, $VBT_{60\text{-}30}$ + RE_{2000}, $VBT_{80\text{-}10}$ + RE_{2000}, and $VBT_{80\text{-}30}$ + RE_{2000}] × time [pre and post]) was also applied on each L-V relationship variable (L_0, v_0, and A_{line}). The Greenhouse-Geisser correction was used when Mauchly's sphericity test was violated and pairwise comparisons were identified using Bonferroni's post hoc corrections. Finally, the Friedman's test was used to compare RE_{2000} time trial between the protocols. The Wilcoxon's signed-rank test with Bonferroni's post hoc corrections was used for pairwise comparisons. The magnitude of the differences was quantified through the standardized mean differences (Hedge's *g* effect size [ES]). The following scale was used to interpret the magnitude of the ES: *trivial* (<0.20), *small* (0.20–0.59), *moderate* (0.60–1.19), *large* (1.20–2.00), and *extremely large* (>2.00) (15). All statistical analyses were performed using the software package SPSS (IBM SPSS version 25.0, Chicago, IL), and statistical significance was set at an alpha level of 0.05.

Results

Descriptive Characteristics of the Velocity-Based Resistance Training Protocols

The ANOVA revealed a significant effect between protocols for the load lifted ($F_{(3,42)} = 64.8$; $p < 0.001$). In addition, the ANOVA conducted on the number of repetitions, fastest velocity, and average velocity revealed a significant main effect for protocol ($F_{(3,42)} \geq 267.3$; $p < 0.001$) and set ($F_{(2,28)} \geq 19.5$; $p < 0.001$). The protocol × set interaction reached statistical significance for the number of repetitions ($F_{(6,84)} = 13.7$; $p < 0.001$) but not for fastest velocity ($F_{(6,84)} = 0.8$; $p = 0.563$) and average velocity ($F_{(6,84)} = 2.6$; $p = 0.061$) (Table 1). The main effects revealed that (a) the absolute load was higher for $VBT_{80\text{-}10}$ (57.3 ± 18.9 kg) and $VBT_{80\text{-}30}$ (56.9 ± 19.5 kg) protocols than for $VBT_{60\text{-}10}$ (43.2 ± 14.4 kg) and $VBT_{60\text{-}30}$ (42.7 ± 14.7 kg) protocols ($p < 0.001$; ES ≥ 0.82), with no significant difference for the same relative loads ($p = 1.000$; ES ≤ 0.03), (b) the number of repetitions was higher for the $VBT_{60\text{-}30}$ protocol, followed by the $VBT_{80\text{-}30}$, $VBT_{60\text{-}10}$, and $VBT_{80\text{-}10}$ protocols ($p < 0.001$; ES ≥ 0.84), (c) the fastest velocity was higher for $VBT_{60\text{-}10}$ and $VBT_{60\text{-}30}$ protocols than for $VBT_{80\text{-}10}$ and $VBT_{80\text{-}30}$ protocols ($p < 0.001$; ES ≥ 7.35), with no significant difference for the same relative loads ($p \geq 0.495$; ES ≤ 0.39), (d) the average velocity was higher for the $VBT_{60\text{-}10}$ protocol, followed by the $VBT_{60\text{-}30}$, $VBT_{80\text{-}10}$, and $VBT_{80\text{-}30}$ protocols ($p < 0.001$; ES ≥ 2.21), (e) the number of repetitions, fastest velocity, and average velocity were progressively reduced with increasing set number ($p \leq 0.001$, 0.045, and 0.004; ES range = 0.26–0.56, 0.07–0.14, and 0.08–0.23, respectively), and (f) the number of repetitions was more markedly reduced for $VBT_{60\text{-}30}$ and $VBT_{80\text{-}30}$ (from −3.6 to −6.5 and from −1.4 to −3.8 average repetitions, respectively) protocols than for $VBT_{60\text{-}10}$ and $VBT_{80\text{-}10}$ (from −0.9 to −2.1 and from −1.0 to −1.6 average repetitions, respectively) protocols with increasing set number.

RE_{2000} Time Trial

Friedman's test revealed a significant effect between the protocols ($\chi^2_{(4,\ N\ =\ 15)} < 0.001$), with longer time trials for $VBT_{60\text{-}30}$ + RE_{2000} and $VBT_{80\text{-}30}$ + RE_{2000} compared with RE_{2000} alone ($p < 0.001$; ES = 0.13 and 0.15, respectively), $VBT_{60\text{-}10}$ + RE_{2000} ($p < 0.001$; ES = 0.10 and 0.12, respectively), and $VBT_{80\text{-}10}$ + RE_{2000} ($p < 0.001$; ES = 0.13 and 0.15, respectively) (Figure 2). No significant differences were observed for the remaining comparisons ($p \geq 0.435$; ES ≤ 0.03).

Changes in Maximal Neuromuscular Capacities

No significant differences were reported for pre L-V profile ($F_{(4,56)} \leq 2.5$; $p \geq 0.091$). The ANOVA did not reveal a significant main effect of protocol for any L-V relationship variable ($F_{(4,56)} \leq 1.9$; $p \geq 0.124$), although a significant main effect of time was observed for L_0 ($F_{(1,14)} = 28.9$; $p < 0.001$) and A_{line} ($F_{(1,14)} = 61.9$; $p < 0.001$) but not for v_0 ($F_{(1,14)} = 3.6$; $p = 0.078$). The protocol × time interaction reached statistical significance for A_{line} ($F_{(4,56)} = 3.1$; $p = 0.041$) but not for L_0 ($F_{(4,56)} = 1.1$; $p = 0.385$) and v_0 ($F_{(4,56)} = 0.3$; $p = 0.902$) (Table 2). The main effects revealed that (a) L_0 and A_{line} were significantly reduced after the different protocols ($p < 0.001$; ES = 0.10 and 0.13, respectively) and (b) A_{line} was reduced to a greater extent for $VBT_{60\text{-}10}$ + RE_{2000} ($p = 0.001$; ES = 0.11), $VBT_{60\text{-}30}$ + RE_{2000} ($p = 0.001$; ES = 0.18), $VBT_{80\text{-}30}$ + RE_{2000} ($p = 0.001$; ES = 0.12), and RE_{2000} ($p = 0.001$; ES = 0.18) protocols than for $VBT_{80\text{-}10}$ + RE_{2000} protocol ($p = 0.065$; ES = 0.05).

Discussion

This study was designed to explore the acute effect of different VBT protocols in terms of loading magnitude (60% vs. 80%

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Table 1

Comparison of the number of repetitions, fastest velocity, and average velocity between velocity-based training (VBT) protocols and sets.*†

						Protocol		Set		Interaction	
Variable	**Set number**	**$VBT_{60\text{-}10}$**	**$VBT_{60\text{-}30}$**	**$VBT_{80\text{-}10}$**	**$VBT_{80\text{-}30}$**	**$F_{(3,42)}$**	***p***	**$F_{(2,28)}$**	***p***	**$F_{(6,84)}$**	***p***
Number of repetitions	1	8.3 ± 1.5	22.9 ± 3.1	5.0 ± 1.6	10.9 ± 2.7	267.3	<0.001	92.9	<0.001	13.7	<0.001
	2	7.4 ± 1.6	19.3 ± 3.3	4.0 ± 1.1	9.5 ± 1.9						
	3	6.1 ± 1.5	16.3 ± 2.3	3.4 ± 1.1	7.1 ± 1.3						
Fastest velocity ($m \cdot s^{-1}$)	1	1.02 ± 0.02	1.04 ± 0.03	0.75 ± 0.01	0.76 ± 0.03	616.1	<0.001	19.5	<0.001	0.8	0.563
	2	1.02 ± 0.02	1.02 ± 0.03	0.74 ± 0.02	0.76 ± 0.03						
	3	1.00 ± 0.02	1.00 ± 0.05	0.73 ± 0.02	0.74 ± 0.02						
Average velocity ($m \cdot s^{-1}$)	1	0.97 ± 0.02	0.91 ± 0.04	0.72 ± 0.01	0.67 ± 0.03	404.0	<0.001	47.5	<0.001	2.6	0.061
	2	0.96 ± 0.02	0.88 ± 0.04	0.71 ± 0.02	0.66 ± 0.04						
	3	0.95 ± 0.02	0.87 ± 0.03	0.70 ± 0.02	0.65 ± 0.04						

*$VBT_{60\text{-}10}$ = VBT protocol against 60% of 1 repetition maximum (1RM) with a velocity loss in the set of 10%; $VBT_{60\text{-}30}$ = VBT protocol against 60% 1RM with a velocity loss in the set of 30%; $VBT_{80\text{-}10}$ = VBT protocol against 80% 1RM with a velocity loss in the set of 10%; $VBT_{80\text{-}30}$ = VBT protocol against 80% 1RM with a velocity loss in the set of 30%; *F* = Snedecor's F.
†Data are presented as means ± *SD*.

1RM) and velocity loss in the set (10% vs. 30%) on RE_{2000} time trial, as well as the behavior of the maximal neuromuscular capacities when RE_{2000} is performed alone or preceded by VBT protocols ($VBT_{60\text{-}10}$ + RE_{2000}, $VBT_{60\text{-}30}$ + RE_{2000}, $VBT_{80\text{-}10}$ + RE_{2000}, and $VBT_{80\text{-}30}$ + RE_{2000}). The main findings revealed that (a) when compared with the control condition, RE_{2000} time trial was compromised by greater velocity loss in the set (30% > 10% velocity loss) but not by the loading magnitude (60% = 80% 1RM), and (b) L_0 and A_{line} were significantly reduced after the different protocols, being the deterioration in A_{line} more accentuated for $VBT_{60\text{-}10}$ + RE_{2000}, $VBT_{60\text{-}30}$ + RE_{2000}, $VBT_{80\text{-}30}$ + RE_{2000}, and RE_{2000} protocols than for $VBT_{80\text{-}10}$ + RE_{2000} protocol. These results suggest that VBT protocols with greater velocity loss in the set (30% vs. 10%) negatively affected subsequent rowing ergometer performance, in line with impairment in A_{line} during the prone bench pull exercise.

Velocity-based resistance training has proliferated as an objective autoregulation method that uses movement velocity to prescribe the intensity (loads) and volume (number of repetitions) during resistance training sessions (34). On the one hand, the individualized L-V relationship was determined daily in this study from the 2-point method based on the rower's readiness to train. This VBT prediction method has been recommended as the most accurate for predicting 1RM during the free-weight prone bench pull exercise (6). Note that both the absolute load and fastest velocity varied between different relative loads but not for the same relative load. However, the wide range of fastest velocities reported in this study for each relative load (from 0.97 to 1.10 $m \cdot second^{-1}$ for 60% 1RM and from 0.72 to 0.85 $m \cdot second^{-1}$ for 80% 1RM) would question the accuracy of previous studies that have prescribed exercise intensity from generalized L-V relationship equations

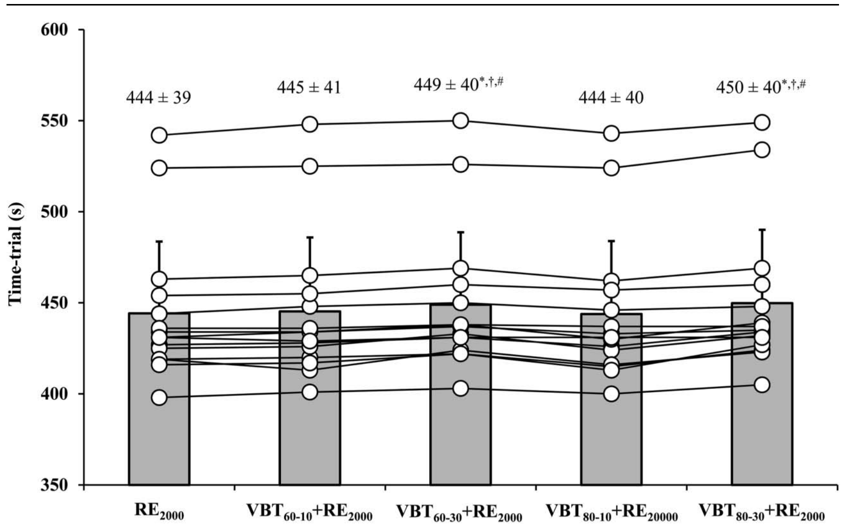

Figure 2. Comparison of the time trial between the protocols. Data are depicted as means and standard deviation, whereas each point represents the data of an individual subject. RE_{2000} = 2000-meter rowing ergometer protocol; $VBT_{60\text{-}10}$ + RE_{2000} = velocity-based training (VBT) protocol against 60% of 1 repetition maximum (1RM) with a velocity loss in the set of 10% followed by RE_{2000}; $VBT_{60\text{-}30}$ + RE_{2000} = VBT protocol against 60% of 1RM with a velocity loss in the set of 30% followed by RE_{2000}; $VBT_{80\text{-}10}$ + RE_{2000} = VBT protocol against 80% of 1RM with a velocity loss in the set of 10% followed by RE_{2000}; $VBT_{80\text{-}30}$ + RE_{2000} = VBT protocol against 80% of 1RM with a velocity loss in the set of 30% followed by RE_{2000}. * = significantly higher than RE_{2000}; † = significantly higher than $VBT_{60\text{-}10}$ + RE_{2000}; # = significantly higher than $VBT_{80\text{-}10}$ + RE_{2000} ($p > 0.05$; Wilcoxon's signed-rank test with Bonferroni's post hoc correction).

Table 2
Comparison of the load-velocity (L-V) relationship variables between protocols and points of measure.*†

L-V relationship variable	Protocol	Pre	Post	Protocol $F_{(4,56)}$	Protocol *p*	Time $F_{(1,14)}$	Time *p*	Interaction $F_{(4,56)}$	Interaction *p*
L_0 (kg)	RE_{2000}	93.6 ± 27.8	90.0 ± 27.5	0.6	0.640	28.9	<0.001	1.1	0.385
	$VBT_{60\text{-}10}$ + RE_{2000}	93.3 ± 27.3	90.8 ± 28.6						
	$VBT_{60\text{-}30}$ + RE_{2000}	95.5 ± 26.4	92.9 ± 27.2						
	$VBT_{80\text{-}10}$ + RE_{2000}	93.4 ± 27.0	92.3 ± 25.5						
	$VBT_{80\text{-}30}$ + RE_{2000}	93.5 ± 27.3	90.7 ± 27.4						
v_0 ($m \cdot s^{-1}$)	RET_{2000}	1.89 ± 0.22	1.84 ± 0.22	0.8	0.488	3.6	0.078	0.3	0.902
	$VBT_{60\text{-}10}$ + RE_{2000}	1.93 ± 0.22	1.90 ± 0.19						
	$VBT_{60\text{-}30}$ + RE_{2000}	1.91 ± 0.24	1.85 ± 0.25						
	$VBT_{80\text{-}10}$ + RE_{2000}	1.87 ± 0.18	1.84 ± 0.25						
	$VBT_{80\text{-}30}$ + RE_{2000}	1.91 ± 0.21	1.89 ± 0.19						
A_{line} ($kg \cdot m \cdot s^{-1}$)	RE_{2000}	89.5 ± 31.5	83.9 ± 30.4	1.9	0.124	61.9	<0.001	3.1	0.041
	$VBT_{60\text{-}10}$ + RE_{2000}	91.5 ± 33.3	88.0 ± 33.3						
	$VBT_{60\text{-}30}$ + RE_{2000}	92.9 ± 34.1	86.9 ± 31.3						
	$VBT_{80\text{-}10}$ + RE_{2000}	88.1 ± 29.5	86.5 ± 29.9						
	$VBT_{80\text{-}30}$ + RE_{2000}	90.7 ± 32.1	87.0 ± 31.5						

*F = Snedecor's F; L_0 = load-axis intercept; v_0 = velocity-axis intercept; A_{line} = area under the L-V relationship line; RE_{2000} = 2000-meter rowing ergometer protocol; $VBT_{60\text{-}10}$ + RE_{2000}= velocity-based training (VBT) protocol against 60% of 1 repetition maximum (1RM) with a velocity loss in the set of 10% followed by RE_{2000}; $VBT_{60\text{-}30}$ + RE_{2000} = VBT protocol against 60% of 1RM with a velocity loss in the set of 30% followed by RE_{2000}; $VBT_{80\text{-}10}$ + RE_{2000} = VBT protocol against 80% of 1RM with a velocity loss in the set of 10% followed by RE_{2000}; $VBT_{80\text{-}30}$ + RE_{2000} = VBT protocol against 80% of 1RM with a velocity loss in the set of 30% followed by RE_{2000}.
†Data are presented as mean ± *SD*s.

(12,21,22,31,33). This finding is in line with García-Ramos et al. (9) who observed that the within-subject variability of the velocity attained at each % 1RM was lower than the between-subject variability for the light-moderate loads. On the other hand, although definitive evidence is lacking (27), a specific magnitude of velocity loss was used to achieve a more homogeneous degree of fatigue across individuals (12,31). For example, in the prone bench pull against 80% 1RM, terminating the set after reaching 10% velocity loss would result in approximately 40–50% of the possible repetitions, whereas a 30% velocity loss would result in approximately 80–90% of the possible repetitions (14). Furthermore, the target velocity loss was set from the fastest velocity of the first set to prevent subjects from reaching or being very close to muscle failure in successive sets. Note that the fastest velocity, and consequently the number of repetitions and average velocity, was reduced with increasing set number because of fatigue.

This study was mainly conducted to answer the question: is rowing ergometer performance compromised to a greater extent by intensity (% 1RM) or volume (velocity loss in the set) of previous resistance training? Partially supporting our first hypothesis, RE_{2000} time trial was compromised by greater velocity loss in the set (30% > 10% velocity loss) but not by the loading magnitude (60% = 80% 1RM). These results are in line with previous studies that have shown that, for the same exercise intensity, greater velocity loss thresholds compromise subsequent endurance performance (23) or increase susceptibility to RT-SEP (34). Several physiological processes have been proposed to explain potential causes related to this phenomenon: (a) impaired neural recruitment patterns; (b) reduced movement efficiency; (c) increased muscle soreness; and (d) reduced muscle glycogen (2,3). However, our findings are in disagreement with Doma et al. (3) who recommended light-load resistance training bouts to further reduce susceptibility to RT-SEP compared with high-load resistance training bouts. By contrast, other studies have shown that, for the same magnitude of velocity loss, not only the mechanical and metabolic fatigue but also the time course of recovery is greater following the light-load session compared with the heavier-load session (22,32). It is possible to assume that although heavy load may result in greater fatigue because of reduced available nonfatigue motor units to recruit, moderate loads involve more mechanical work and physiological stress, which may also induce greater levels of fatigue (22). Note that in this study, the same velocity loss threshold was set in the set, but the mechanical work was not equalized (i.e., subjects completed greater number of repetitions at 60% than 80% 1RM for the same velocity loss). Therefore, rowing coaches should be especially aware of the impairment in rowing ergometer performance when is preceded by VBT protocols with a high velocity loss in the set.

The determination of the force-velocity or L-V relationship has been recommended to selectively assess the effects of fatigue upon the distinctive maximal neuromuscular capacities during multijoint tasks (8). Rejecting our second hypothesis, a decrement in A_{line} was observed because of the decrease in L_0 and not v_0 after the different protocols. Discordant with our findings, previous research found a reduction in P_{max} after strenuous exercise protocols because of greater deterioration of v_0 than F_0 (8,23). For example, Pérez-Castilla et al. (23) observed that F_0 tended to decrease when VBT was performed alone or preceded by a sprint interval training but increase when high-intensity interval training was followed by VBT. However, although all previous studies determined the force-velocity relationship during ballistic exercises (e.g., countermovement jump or bench press throw), the free-weight prone bench pull exercise was used in this study to determine the L-V relationship. Therefore, because ballistic performance is highly dependent on v_0 capacity (1) and the lower-limb (squat jump) and upper-limb (prone bench pull) force-velocity profile is more force-oriented in rowers (29), it is plausible that the impairment of maximal neuromuscular capacities may depend on exercise selection also. Moreover, the deterioration in A_{line} was more accentuated for $VBT_{60\text{-}10}$ + RE_{2000}, $VBT_{60\text{-}30}$ + RE_{2000}, $VBT_{80\text{-}30}$ + RE_{2000}, and RE_{2000} protocols than for $VBT_{80\text{-}10}$ + RE_{2000} protocol. It seems that VBT inducing higher level of fatigue (i.e., a greater number of repetitions completed in each set) can impair the low-velocity region of the L-V relationship. By contrast, the lower A_{line} impairment for $VBT_{80\text{-}10}$ + RE_{2000} compared with RE_{2000} alone might suggest that rowing efficiency may improve after a nonfatiguing VBT session. Collectively, our novel findings provide useful insight into the use of L-V relationship to assess the effects of fatigue upon the maximal neuromuscular capacities.

Several limitations need to be acknowledged when interpreting the findings from this study. First, because muscle damage and attenuation in muscle function may depend on training background (2), the fact of including male competitive rowers of different age categories (cadet, junior, and senior) may have limited the generalizability of the current findings. Second, although the force-velocity relationship remains linear under fatigue conditions (8), the accuracy of the L-V relationship variables may be affected using only 2 loads for modeling. Future studies should examine the feasibility of the 2-point method to monitor fatigue-associated changes in muscle mechanical capacities.

Practical Applications

Velocity-based resistance training is an objective method that allows the provision of real-time velocity feedback, and the individualization of exercise intensity and volume based on an individual's daily readiness to train. Practitioners implementing this contemporary methodology should be mindful that rowing ergometer performance is compromised when it is immediately preceded by VBT with high velocity loss thresholds in the set (≥30% velocity loss). Instead, exercise intensity (% 1RM) used to set VBT does not appear to affect subsequent rowing ergometer performance (60% = 80% 1RM). Furthermore, the L-V relationship can be used to selectively assess the effects of fatigue upon the distinctive maximal neuromuscular capacities during multijoint tasks. Our results indicate that L_0 and A_{line} obtained during the prone bench pull exercise are meaningfully reduced after the different protocols, being the deterioration in A_{line} more accentuated for $VBT_{60\text{-}10}$ + RE_{2000}, $VBT_{60\text{-}30}$ + RE_{2000}, $VBT_{80\text{-}30}$ + RE_{2000}, and RE_{2000} protocols than for $VBT_{80\text{-}10}$ + RE_{2000} protocol. This finding should be of special interest to practitioners because managing residual fatigue could be of paramount importance to reducing susceptibility to RT-SEP.

Acknowledgments

The authors thank all the subjects who selflessly participated in the study. The results of this study do not constitute an endorsement of the product by the authors or the National Strength and Conditioning Association. There is no conflict of interest. There are no professional relationships with companies or manufacturers who will benefit from the results of this study for each author.

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